The documentation and process conversion measures necessary to comply with this revision shall be completed by 16 April 2012.

INCH-POUND

MIL-PRF-19500/518D
w/AMENDMENT 1
2 March 2012
SUPERSEDING
MIL-PRF-19500/518D
11 December 2007

PERFORMANCE SPECIFICATION SHEET

SEMICONDUCTOR DEVICE, TRANSISTOR, NPN, SILICON, POWER,
TYPE 2N3766, 2N3767, JAN, JANTX, AND JANTXV

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-PRF-19500.

1. SCOPE

1.1 Scope. This specification covers the performance requirements for NPN silicon, power transistors. Three levels of product assurance are provided for each device type as specified in MIL-PRF-19500.

1.2 Physical dimensions. See figure 1 (similar to TO-66).

1.3 Maximum ratings. Unless otherwise specified, T_C = +25°C.

Type	P_T (1)	V_{CBO}	V_{CEO}	V_{EBO}	I_B	I_C	T_J and T_{STG}	$R_{\theta JC}$
	W	V dc	V dc	V dc	A dc	A dc	°C	°C/W
2N3766	25	80	60	6	2	4	-65 to +200	7
2N3767	25	100	80	6	2	4	-65 to +200	7

(1) Between T_C = +25°C and T_C = +200°C, linear derating factor (average) = 143 m/W°C.

* Comments, suggestions, or questions on this document should be addressed to DLA Land and Maritime, ATTN: VAC, P.O. Box 3990, Columbus, OH 43218-3990, or emailed to Semiconductor@dla.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.daps.dla.mil/.

AMSC N/A

FSC 5961

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1.4 Primary electrical characteristics.

Type	h_{FE2} (1)		h_{FE3} (1)		C_{obo}		$\lvert h_{fe} \rvert$		$V_{BE(sat)}$	
	V_{CE} = 5 V dc I_C = 500 mA dc		V_{CE} = 10 V dc I_C = 1 A dc		V_{CB} = 10 V dc I_E = 0 0.1 MHz ≤ f ≤ 1 MHz		V_{CE} = 10 V dc I_C = 500 mA dc f = 10 MHz		I_C = 1 A dc I_B = 0.1 A dc	
	Min	Max	Min	Max	Min PF	Max pF	Min	Max	Min V dc	Max V dc
2N3766	40	160	20			50	1	8		1.5
2N3767	40	160	20			50	1	8		1.5

Type	$V_{CE(sat)1}$		$V_{CE(sat)2}$		Switching			
	I_C = 1 A dc I_B = 0.1 A dc		I_C = 500 mA dc I_B = 50 mA dc		t_{on}		t_{off}	
					(see table I and figure 2 herein)			
	Min	Max	Min	Max	Min µs	Max µs	Min µs	Max µs
2N3766		2.5		1.0		0.25		2.5
2N3767		2.5		1.0		0.25		2.5

(1) Pulsed (see 4.5.1).

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3, 4, or 5 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3, 4, or 5 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-PRF-19500 - Semiconductor Devices, General Specification for.

DEPARTMENT OF DEFENSE STANDARDS

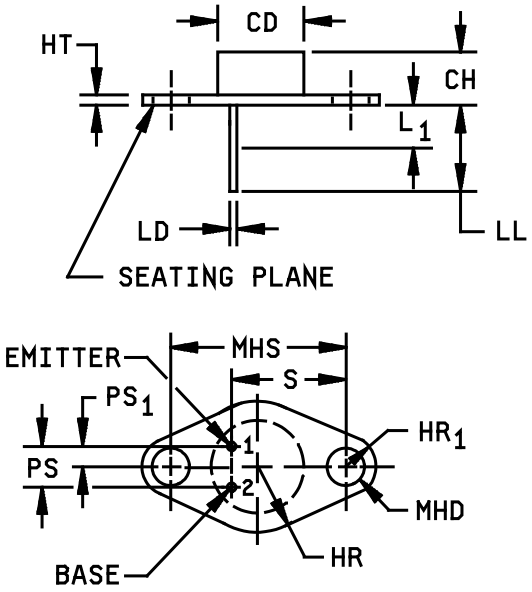
MIL-STD-750 - Test Methods for Semiconductor Devices.

* (Copies of these documents are available online at https://assist.daps.dla.mil/quicksearch/ or https://assist.daps.dla.mil/ or from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

* 2.3 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

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Symbol	Dimensions				Notes
	Inches		Millimeters		
	Min	Max	Min	Max	
CD		.620		15.76	
CH	.250	.340	6.35	8.64	
HR		.350		8.89	4
HT	.050	.075	1.27	1.91	
HR_1	.115	.145	2.92	3.68	4
LD	.028	.034	0.71	0.86	4, 6
LL	.360	.500	9.14	12.70	
L_1		.050		1.27	6
MHD	.142	.152	3.61	3.86	4
MHS	.958	.962	24.33	24.43	
PS	.190	.210	4.83	5.33	3
PS_1	.093	.107	2.36	2.72	3
S	.570	.590	14.48	14.99	



NOTES:
1. Dimensions are in inches.
2. Millimeters are given for general information only.
3. These dimensions should be measured at points .050 inch (1.27 mm) +.005 inch (0.13 mm) -.000 inch (0.00 mm) below seating plane. When gauge is not used, measurement will be made at the seating plane.
4. Two places.
5. The seating plane of the header shall be flat within .001 inch (0.03 mm) concave to .004 inch (0.10 mm) convex inside a .930 inch (23.62 mm) diameter circle on the center of the header and flat within .001 inch (0.03 mm) concave to .006 inch (0.15 mm) convex overall.
6. Lead diameter shall not exceed twice LD within L_1.
7. Lead number 1 is the emitter, lead 2 is the base, case is the collector.
8. In accordance with ASME Y14.5M, diameters are equivalent to ϕx symbology.

FIGURE 1. Physical dimensions (similar to TO-66).

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3. REQUIREMENTS

3.1 General. The individual item requirements shall be as specified in MIL-PRF-19500 and as modified herein.

3.2 Qualification. Devices furnished under this specification shall be products that are manufactured by a manufacturer authorized by the qualifying activity for listing on the applicable qualified manufacturer's list (QML) before contract award (see 4.2 and 6.3).

3.3 Abbreviations, symbols, and definitions. Abbreviations, symbols, and definitions used herein shall be as specified in MIL-PRF-19500.

3.4 Interface requirements and physical dimensions. The interface requirements and physical dimensions shall be as specified in MIL-PRF-19500 and on figure 1 herein.

* 3.4.1 Lead finish. Unless otherwise specified, lead finish shall be solderable in accordance with MIL-PRF-19500, MIL-STD-750, and herein. Where a choice of lead finish is desired, it shall be specified in the acquisition document (see 6.2).

3.5 Marking. Marking shall be in accordance with MIL-PRF-19500.

* 3.6 Electrical performance characteristics. Unless otherwise specified herein, the electrical performance characteristics are as specified in 1.3, 1.4, and table I.

* 3.7 Electrical test requirements. The electrical test requirements shall be as specified in table I.

3.8 Workmanship. Semiconductor devices shall be processed in such a manner as to be uniform in quality and shall be free from other defects that will affect life, serviceability, or appearance.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.2).

b. Screening (see 4.3)

c. Conformance inspection (see 4.4 and tables I and II).

4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-PRF-19500 and as specified herein.

4.2.1 Group E qualification. Group E inspection shall be performed for qualification or re-qualification only. In case qualification was awarded to a prior revision of the specification sheet that did not request the performance of table II tests, the tests specified in table II herein that were not performed in the prior revision shall be performed on the first inspection lot of this revision to maintain qualification.

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4.3 Screening (JANTX and JANTXV levels only). Screening shall be in accordance with table E-IV of MIL-PRF-19500 and as specified herein. The following measurements shall be made in accordance with table I herein. Devices that exceed the limits of table I herein shall not be acceptable.

Screen (see table E-IV of MIL-PRF-19500)	Measurement JANTX and JANTXV levels
(1) 3c	Thermal impedance (see 4.3.2).
11	I_{CEX1} and h_{FE2}.
12	See 4.3.1.
13	Subgroup 2 of table I herein. ΔI_{CEX1} = 100 percent of initial value or 1 μA dc, whichever is greater; Δh_{FE2} = 25 percent of initial value.

(1) Shall be performed anytime after temperature cycling, screen 3a; and does not need to be repeated in screening requirements.

4.3.1 Power burn-in conditions. Power burn-in conditions are as follows: T_J = +187.5°C ±12.5°C, $V_{CB} \geq$ 10 V dc, $T_A \leq$ +35°C.

* 4.3.2 Thermal impedance. The thermal impedance measurements shall be performed in accordance with method 3131 of MIL-STD-750 using the guidelines in that method for determining I_M, I_H, t_H, t_{SW}, (and V_H where appropriate). Measurement delay time (t_{MD}) = 70 μs max.

* 4.4 Conformance inspection. Conformance inspection shall be in accordance with MIL-PRF-19500, and as specified herein. If alternate screening is being performed in accordance with MIL-PRF-19500, a sample of screened devices shall be submitted to and pass the requirements of subgroups 1 and 2, of table I herein, inspection only (table E-VIb, group B, subgroup 1 is not required to be performed again if group B has already been satisfied in accordance with 4.4.2).

* 4.4.1 Group A inspection. Group A inspection shall be conducted in accordance with table E-IV of MIL-PRF-19500, and table I herein.

* 4.4.2 Group B inspection. Group B inspection shall be conducted in accordance with the conditions specified for subgroup testing in table E-VIc of MIL-PRF-19500. Separate samples may be used for each step. Electrical measurements (end-points) shall be in accordance with table I, subgroup 2 herein.

Step	Method	Condition
1	1039	Steady-state life: Test condition B, 340 hours, $V_{CB} \geq$ 10 V dc, T_J = 175°C min. No heat sink or forced-air cooling on the devices shall be permitted. n = 45 devices, c = 0.
2	1039	The steady-state life test of step 1 shall be extended to 1,000 hours for each die design. Samples shall be selected from a wafer lot every 12 months of wafer production; however, group B shall not be required more than once for any single wafer lot. n = 45, c = 0.
3	1032	High-temperature life (non-operating), T_A = +200°C. n = 22, c = 0

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4.4.3 Group C inspection. Group C inspection shall be conducted in accordance with the conditions specified for subgroup testing in table E-VII of MIL-PRF-19500. Electrical measurements (end-points) shall be in accordance with table I, subgroup 2 herein.

4.4.3.1 Group C inspection, table E-VII (JAN, JANTX, and JANTXV) of MIL-PRF-19500.

Subgroup	Method	Condition
C2	2036	Test condition A, weight = 10 lbs, T = 15 seconds.
C5	3131	Thermal resistance, see 4.3.2, $R_{\theta JC(max)}$ = 7°C/W.
C6		Not applicable

4.5 Methods of inspection. Methods of inspection shall be as specified in the appropriate tables and as follows.

4.5.1 Pulse measurements. Conditions for pulse measurement shall be as specified in section 4 of MIL-STD-750.

4.5.2 Coil selection for safe operating area (SOA) tests. In selecting coils for use in clamped and unclamped inductive SOA tests, prime consideration should be given to the commercially available coil. However, due to the extreme critical nature of the coil in these circuits and wide tolerance of some commercially available coils (+100 percent, -50 percent), it shall be the semiconductor manufacturer's responsibility to prove upon request compliance or equivalency of any coil used (commercial or in-plant designed) to be within (+20 percent, -10 percent) of the specified inductance at the rated current and dc resistance.

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TABLE I. Group A inspection.

Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 1 2/						
Visual and mechanical 3/ examination	2071					
Solderability 3/	2026					
Resistance to 3/ 4/ solvent	1022					
Temp cycling 3/	1051	Test condition C, 25 cycles.				
Hermetic seal	1071					
Fine leak Gross leak						
Electrical measurements		Group A, subgroup 2				
Bond strength 3/	2037					
Subgroup 2						
Thermal impedance 5/	3131	See 4.3.2	$Z_{\theta JC}$			°C/W
Collector to emitter breakdown voltage	3011	Bias condition D, I_C = 100 mA dc, pulsed (see 4.5.1)	$V_{(BR)CEO}$			
2N3766 2N3767				60 80		V dc V dc
Collector to emitter cutoff current 2N3766 2N3767	3041	Bias condition D, V_{CE} = 60 V dc V_{CE} = 80 V dc	I_{CEO}		500	µA dc
Emitter-base cutoff current	3061	Bias condition D, V_{EB} = 6 V dc	I_{EBO}		500	µA dc
Collector to emitter cutoff current 2N3766 2N3767	3041	Bias condition A, V_{BE} = 1.5 V dc, V_{CE} = 80 V dc V_{CE} = 100 V dc	I_{CEX1}		 10 10	 µA dc µA dc
Collector to base cutoff current 2N3766 2N3767	3036	Bias condition D, V_{CB} = 80 V dc V_{CB} = 100 V dc	I_{CBO}		 10 10	 µA dc µA dc

See footnotes at end of table.

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TABLE I. Group A inspection - Continued.

Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 2 - Continued						
Base to emitter non-saturated voltage	3066	Test condition B, V_{CE} = 10 V dc, I_C = 1.0 A dc, pulsed (see 4.5.1)	V_{BE}		1.5	V dc
Base to emitter saturated voltage	3066	Test condition A, I_C = 1.0 A dc, I_B = 0.1 A dc, pulsed (see 4.5.1)	$V_{BE(sat)1}$		1.5	V dc
Collector to emitter saturated voltage	3071	I_C = 1.0 A dc, I_B = 0.1 A dc, pulsed (see 4.5.1)	$V_{CE(sat)1}$		2.5	V dc
Collector to emitter saturated voltage	3071	I_C = 0.5 A dc, I_B = 0.05 A dc, pulsed (see 4.5.1)	$V_{CE(sat)2}$		1.0	V dc
Forward-current transfer ratio	3076	V_{CE} = 5 V dc, I_C = 50 mA dc, pulsed (see 4.5.1)	h_{FE1}	30		
Forward-current transfer ratio	3076	V_{CE} = 5 V dc, I_C = 500 mA dc, pulsed (see 4.5.1)	h_{FE2}	40	160	
Forward-current transfer ratio	3076	V_{CE} = 10 V dc, I_C = 1 A dc, pulsed (see 4.5.1)	h_{FE3}	20		
Subgroup 3						
High-temperature operation:		T_A = +150°C				
Collector to emitter cutoff current	3041	Bias condition A, V_{BE} = 1.5 V dc	I_{CEX2}			
2N3766 2N3767		V_{CE} = 50 V dc V_{CE} = 70 V dc			1.0 1.0	mA dc mA dc
Low-temperature operation:		T_A = -55°C				
Forward-current transfer ratio	3086	V_{CE} = 5.0 V dc, I_C = 0.5 A dc, pulsed (see 4.5.1)	h_{FE4}	13		

See footnotes at end of table.

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TABLE I. Group A inspection - Continued.

Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 4						
Pulse response:	3251	Test condition A, except test circuit and pulse requirements in accordance with figure 2 herein.				
Turn-on time		V_{CC} = 30 V dc, I_C = 0.5 A dc, I_{B1} = 0.05 A dc	t_{on}		0.25	µs
Turn-off time		V_{CC} = 30 V dc, I_C = 0.5 A dc, I_{B1} = I_{B2} = .05 A dc	t_{off}		2.5	µs
Magnitude of common emitter small-signal short-circuit forward-current transfer ratio	3306	V_{CE} = 10 V dc, I_C = 500 mA dc, f = 10 MHz	$\lvert h_{fe} \rvert$	1	8	
Open capacitance (open circuit)	3236	V_{CB} = 10 V dc, I_E = 0, 0.1 MHz ≤ f ≤ 1.0 MHz	C_{obo}		50	pF
Subgroup 5						
Safe operating area (continuous dc)	3051	T_C = +25°C, power application t = 1 s, 1 cycle, (see figure 3)				
Test 1		V_{CE} = 6.25 V dc, I_C = 4 A dc				
Test 2		V_{CE} = 20 V dc, I_C = 1.25 A dc				
Test 3						
2N3766		V_{CE} = 50 V dc, I_C = 150 mA dc				
2N3767		V_{CE} = 65 V dc, I_C = 150 mA dc				

See footnotes at end of table.

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TABLE I. Group A inspection - Continued.

Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 5 - Continued						
Safe operating area (clamped switching)	3053	Load condition B, T_A = +25°C, V_{CC} = 20 V dc, $t_p \approx$ 1.5 ms, R_{BB1} = 5 Ω, (Vary to obtain I_C), duty cycle ≤ 5 percent, $I_C \approx$ 4 A dc, R_{BB2} = 100 Ω, V_{BB2} = 1.5 V dc, $V_{BB1} \approx$ 10 V dc, L = 5 mH, at 4A, R of inductor = .5 Ω, and R_{LOAD} = R of inductor. 6/				
2N3766 2N3767		Clamp voltage = 60 V dc Clamp voltage = 80 V dc Device fails if clamp voltage is not reached.				
End point electrical measurements		See table I, subgroup 2 herein, I_{CEX1} and h_{FE2} electrical tests				
Subgroups 6 and 7						
Not applicable						

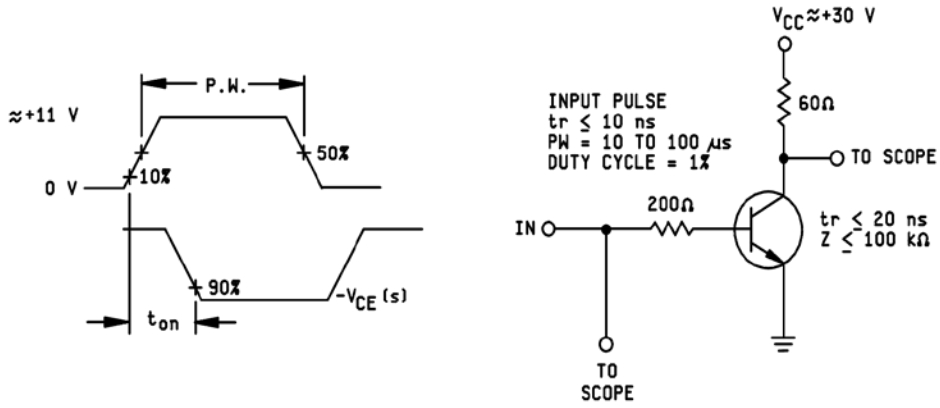
1/ For sampling plan, see MIL-PRF-19500.
2/ For resubmission of failed subgroup A1, double the sample size of the failed test or sequence of tests.
3/ Separate samples may be used.
4/ Not required for laser marked devices.
5/ This test required for the following end-point measurements only:
Group B, all steps (JAN, JANTX, and JANTXV).
Group C, subgroup 2.
Group E, subgroup 1.
6/ L = 5 mH (2 each Essex Stancor C-2688 in parallel 1A, 0.5 ohm, or equivalent (see 4.5.2).

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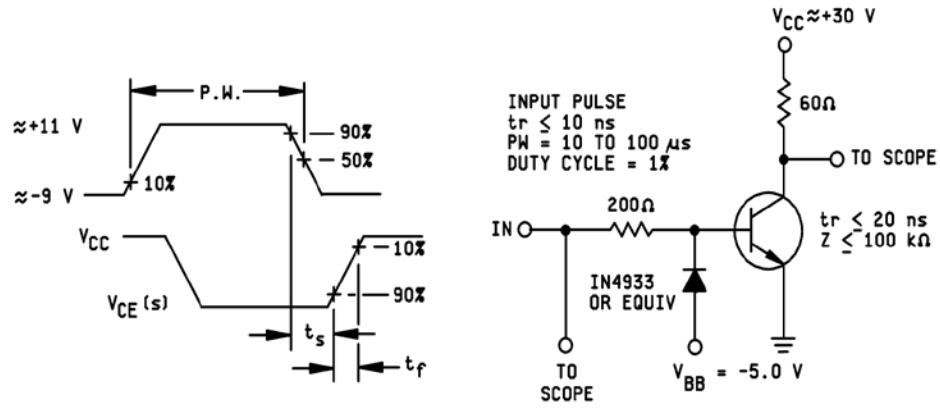
* TABLE II. Group E inspection (all quality levels) for qualification only.

Inspection	MIL-STD-750		Sample plan
	Method	Conditions	
Subgroup 1			45 devices, c = 0
Temperature cycling	1051	500 cycles, condition D	
Hermetic seal Fine leak Gross leak	1071		
Electrical measurements		See table I, subgroup 2.	
Subgroup 2			45 devices, c = 0
Steady-state dc blocking life	1039 or 1049	Condition A, 500 hours	
Electrical measurements		See table I, subgroup 2.	
Subgroup 4			Sample size N/A
Thermal impedance curves		See table E-IX of MIL-PRF-19500, group E, subgroup 4 .	
Subgroup 6			
ESD	1020		
Subgroup 8			45 devices, c = 0
Reverse stability	1033	Condition A.	

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TURN-ON (t_{on}) TIME TEST CIRCUIT



TURN-OFF (t_{off}) TIME TEST CIRCUIT

FIGURE 2. Switching time test circuit.

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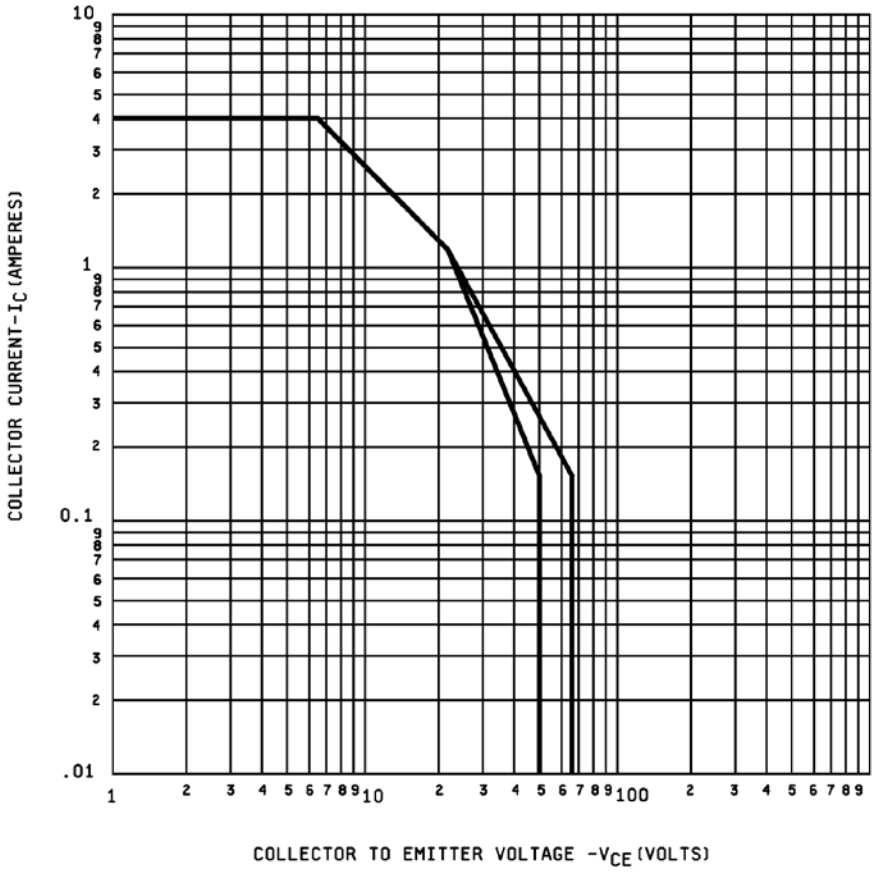


FIGURE 3. Maximum safe operating area graph (continuous dc) for types 2N3766 and 2N3767.

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5. PACKAGING

* 5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the Military Service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory. The notes specified in MIL-PRF-19500 are applicable to this specification.)

6.1 Intended use. Semiconductors conforming to this specification are intended for original equipment design applications and logistic support of existing equipment.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Packaging requirements (see 5.1).

c. Lead finish (see 3.4.1).

d. Product assurance level and type designator.

* 6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Manufacturers List (QML 19500) whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from DLA Land and Maritime, ATTN: VQE, P.O. Box 3990, Columbus, OH 43218-3990 or e-mail vqe.chief@dla.mil. An online listing of products qualified to this specification may be found in the Qualified Products Database (QPD) at https://assist.daps.dla.mil .

* 6.4 Changes from previous issue. The margins of this specification are marked with asterisks to indicate modification generated by this amendment. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations.

Custodians:
Army - CR
Air Force - 85
DLA – CC

Preparing activity:
DLA - CC

(Project 5961-2011-085)

Review activities:
Army- AR
Air Force - 19, 99

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